

1.3 - Multi-Dealer Competition and Systemic Performative Instability

REFLEX project - math-theory

STATUS: DONE (Priority 3, derivation/theorem). This document discharges the *mathematical* content of methodology target **1.3**. It lifts the single-dealer closed forms of **01-analytic-stability-boundary** to N competing dealers who share a common informed-flow pool, and derives the performatively stable Nash equilibrium (PSNE) boundary

$$\text{stable} \iff m_N = N_{\text{eff}} \frac{\varepsilon \beta}{\gamma} < 1 \iff \varepsilon < \frac{\gamma}{N_{\text{eff}} \beta}, \quad N_{\text{eff}} = 1 + \kappa (N - 1),$$

which at full toxic spillover $\kappa = 1$ ($N_{\text{eff}} = N$) is the target $\varepsilon < \gamma/(N\beta)$, and at $\kappa = 0$ recovers the single-dealer boundary of 1.1 applied to N decoupled dealers. Every constant is one of 1.1's closed forms; the lift needs only the N -dealer linear algebra and *no new estimation*.

One-line thesis. A single dealer is stable iff $m_1 = \varepsilon\beta/\gamma < 1$. Put N of them in a market that shares one informed-flow pool and the destabilising feedback no longer acts through each dealer's own re-tightening alone: when *any* dealer tightens it summons toxic flow that picks off *all* of them, so the joint best-response map is a rank-one common-mode coupling whose unstable eigenvalue is N times larger. The market crosses into instability at $\varepsilon = \gamma/(N\beta)$ – a factor N before any individual dealer would. Competition manufactures a synchronised cobweb more fragile than any dealer in isolation. That is the systemic-risk result.

1 Carry-over and new notation

From **01-analytic-stability-boundary** we reuse the single-dealer one-step objective at frozen toxic level T and its three closed-form constants:

$$\begin{aligned} J(h; T) &= P \left[h A e^{-kh} \rho + hT - \psi T - w(h - h_{\text{ref}})^2 - \lambda_q \right], & P &= \text{pnl_scale}, \\ \gamma &= -\partial_{hh} J = P \left[2w + A \rho k e^{-kh} (2 - kh) + \lambda_q \right] & & \text{(strong convexity),} \\ \beta &= |\partial_{hT} J| = P & & \text{(joint smoothness),} \\ \varepsilon &= |d\tau/dh| = \rho \bar{g} \alpha f I c_t e^{-c_t h} & & \text{(distribution sensitivity),} \\ m_1 &= \varepsilon\beta/\gamma & & \text{(single-dealer modulus, 1.1 §3.4).} \end{aligned}$$

Symbol	Reading	Definition
N	number of competing dealers	proposed <code>clients.n_dealers</code>
h_i	half-spread quoted by dealer i	per-dealer policy coordinate
$\mathbf{h} = (h_1, \dots, h_N)$	joint quote profile	strategy profile in $\mathbb{R}^N$
h_i^{dep}	deployed spread of dealer i	§2, A3'
τ_i	toxic notional hitting dealer i	§2
κ	cross-dealer toxic spillover, $\in [0, 1]$	proposed <code>clients.toxic_spillover</code>
T_i	frozen toxic level of dealer i , $= \tau_i(\mathbf{h}^{\text{dep}})$	§2
BR_i	dealer i 's frozen-environment best response	§3
J_{BR}	Jacobian of $\text{BR} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ at the PSNE	§3
N_{eff}	effective dealer count, $= 1 + \kappa(N - 1)$	§4
m_N	joint modulus, $= N_{\text{eff}} m_1 = \rho(J_{\text{BR}})$	§4
γ_{joint}	joint curvature, $= \gamma - N_{\text{eff}} \varepsilon \beta$	§6
N_c	critical dealer count, $= \gamma/(\beta \varepsilon) = 1/m_1$	§8
c	aggregate spillover in the scaling $\kappa = c/N$	§7

2 Setup: N symmetric dealers sharing one informed-flow pool

We extend the single-bond, zero-skew analysis of 1.1 to N dealers quoting the same bond simultaneously, at the level of each dealer's central half-spread h_i .

Assumption 1 (symmetric dealers, single bond, zero skew). *All N dealers share the microstructure constants $(A, k, w, h_{\text{ref}}, P, \lambda_q)$ and the toxic parameters $(I_b, I, \alpha, f, c_t, \sigma_s)$ of 1.1. One bond, skew = 0.*

Assumption 2 (reference state). *$\gamma, \bar{g}, \psi$ are evaluated at the common probe reference state s_0 with $q_0 \approx 0$, mispricing g , liquidity ratio ρ – identical to 1.1.*

Assumption 3 (frozen-environment, simultaneous best response). *Within one deployment round the operator conditions on a frozen `policy_summary` of the whole deployed profile $\mathbf{h}^{\text{dep}}$. Each dealer i sees the toxic environment frozen at $T_i = \tau_i(\mathbf{h}^{\text{dep}})$ while its own benign channel responds to its candidate h_i ; all dealers redeploy together (simultaneous RRM, the multiplayer setting of Brown et al. 2021 and Narang et al. 2022).*

Assumption 4 (saturation non-binding). *The informed-flow cap is slack at the operating point; the uncapped form is carried.*

Assumption 5 (own-franchise benign flow). *Each dealer's uninformed notional $A e^{-kh_i} \rho$ responds to its own spread only; the competitive coupling is carried entirely by the toxic channel (§3). Competitive splitting of benign flow only adds own- h curvature (raises γ , stabilising) and is out of scope (§11).*

3 Coupled toxic flow and the joint best-response map

Shared informed-flow pool. With N dealers the informed pool is shared: a tighter market – any dealer tightening – raises the venue's attractiveness to informed flow, increasing the toxic

notional routed at *every* dealer. We model

$$\tau_i(\mathbf{h}^{\text{dep}}) = \rho \bar{g} \left[I_b + \alpha f I \left(e^{-c_t h_i^{\text{dep}}} + \kappa \sum_{j \neq i} e^{-c_t h_j^{\text{dep}}} \right) \right], \quad \kappa \in [0, 1]. \quad (1)$$

$\kappa = 0$ gives N independent single-dealer problems; $\kappa = 1$ is full spillover (the informed pool tracks the aggregate attractiveness $\sum_j e^{-c_t h_j^{\text{dep}}}$, the “shared induced distribution” of Brown et al. 2021). At the symmetric point $h_j^{\text{dep}} = h^*$,

$$\partial_{h_i^{\text{dep}}} \tau_i = -\varepsilon, \quad \partial_{h_j^{\text{dep}}} \tau_i = -\kappa \varepsilon \ (j \neq i), \quad \varepsilon = \rho \bar{g} \alpha f I c_t e^{-c_t h^*}.$$

Per-dealer objective is 1.1’s. Under A1’/A5’, dealer i ’s objective at frozen T_i is 1.1’s J with $h \rightarrow h_i$, $T \rightarrow T_i$; competitor spreads enter *only* through the scalar T_i . Hence $\partial_{h_i h_i} J_i = -\gamma$ and $\partial_{h_i T_i} J_i = +\beta = P$ exactly as in 1.1, and the constant-in- h_i adverse term $-\psi T_i$ drops out of the boundary (it drives the level / echo-chamber gap, §11).

The map. Each dealer best-responds, $\text{BR}_i(\mathbf{h}^{\text{dep}}) = \arg \max_{h_i} J_i(h_i; \tau_i(\mathbf{h}^{\text{dep}}))$, with FOC $F_i = P[A\rho e^{-kh_i}(1 - kh_i) + T_i - 2w(h_i - h_{\text{ref}})] = 0$. The simultaneous-RRM iteration is $\mathbf{h}^{t+1} = \text{BR}(\mathbf{h}^t)$; a PSNE $\mathbf{h}^*$ is a fixed point, and the loop contracts iff $\rho(J_{\text{BR}}) < 1$.

Jacobian (implicit-function theorem). Differentiating $F_i(\text{BR}_i, \tau_i) = 0$ in h_j^{dep} with $F_{i, h_i} = -\gamma$, $F_{i, T_i} = +\beta$ and the slopes above,

$$\frac{\partial \text{BR}_i}{\partial h_i^{\text{dep}}} = -\frac{\beta \varepsilon}{\gamma} = -m_1 \text{ (diagonal)}, \quad \frac{\partial \text{BR}_i}{\partial h_j^{\text{dep}}} = -\kappa \frac{\beta \varepsilon}{\gamma} = -\kappa m_1 \ (j \neq i), \quad (2)$$

so at the symmetric PSNE

$$J_{\text{BR}} = -m_1 [(1 - \kappa) I + \kappa \mathbf{1}\mathbf{1}^\top], \quad m_1 = \varepsilon \beta / \gamma. \quad (3)$$

J_{BR} is symmetric (consequence of the symmetric shared pool), so $\rho(J_{\text{BR}})$ equals its operator 2-norm and the contraction is a genuine Euclidean contraction.

4 Spectrum, the PSNE boundary, and the headline theorem

The all-ones matrix $\mathbf{1}\mathbf{1}^\top$ has eigenvalue N (eigenvector $\mathbf{1}$, the *common mode*: all dealers move together) and 0 with multiplicity $N - 1$ (the *differential modes*). Hence (3) has eigenvalues

$$\begin{aligned} \lambda_+ &= -m_1 (1 + \kappa(N - 1)) = -m_1 N_{\text{eff}} && \text{(common mode),} \\ \lambda_- &= -m_1 (1 - \kappa) && \text{(differential, mult. } N - 1). \end{aligned}$$

With $N_{\text{eff}} := 1 + \kappa(N - 1) \geq 1 \geq 1 - \kappa$, the common mode dominates and $\rho(J_{\text{BR}}) = m_1 N_{\text{eff}} =: m_N$.

Theorem 1 (PSNE stability boundary). *For N symmetric dealers sharing a toxic pool with spillover $\kappa \in [0, 1]$, the simultaneous-RRM loop contracts to the PSNE iff*

$$m_N = N_{\text{eff}} \frac{\varepsilon \beta}{\gamma} < 1 \iff \varepsilon < \frac{\gamma}{N_{\text{eff}} \beta}, \quad N_{\text{eff}} = 1 + \kappa(N - 1).$$

In particular at full spillover $\kappa = 1$ ($N_{\text{eff}} = N$) this is $\varepsilon < \gamma/(N\beta)$, with market-wide instability at $N\varepsilon\beta/\gamma = 1$; at $\kappa = 0$ it reduces to the single-dealer $\varepsilon < \gamma/\beta$ applied to each of the N decoupled dealers.

Proof. BR is C^1 near the symmetric fixed point (J_i is smooth and strongly concave in h_i by $\gamma > 0$, so BR_i is a smooth implicit function). The iteration contracts near $\mathbf{h}^*$ iff $\rho(J_{\text{BR}}) < 1$ (Ostrowski/Banach for the linearised map); by the spectrum above $\rho(J_{\text{BR}}) = m_1 N_{\text{eff}}$. Substituting $m_1 = \varepsilon\beta/\gamma$ gives the boundary. Symmetry of J_{BR} makes ρ the 2-norm, so $m_N < 1$ is a strict Euclidean contraction. $\square$

Every entry of J_{BR} carries the *same* single-dealer feedback $-m_1$, but the common-mode eigenvalue sums N_{eff} of them in phase: the unstable direction is the synchronised market-wide tightening, not any one dealer’s own loop. Moreover $|\lambda_-| = m_1(1 - \kappa) \leq m_1 < m_N$, so the differential modes are damped *more* strongly than a lone dealer – instability here is intrinsically systemic.

5 Existence and uniqueness of the PSNE

Each $h_i \in [0, \text{max_half_spread}]$, so the joint space B is a compact convex box. Under A1’ each $J_i(\cdot; T_i)$ is strongly concave in h_i ($\gamma > 0$) and τ_i is continuous, so $\text{BR} : B \rightarrow B$ is a continuous self-map of a nonempty compact convex set; by **Brouwer** a PSNE exists (the market-making instance of Brown et al. 2021). When $m_N < 1$, BR is a 2-norm contraction, so the PSNE is **unique** and globally attracting on B (Banach). By symmetry it solves the scalar equation

$$A\rho e^{-kh^*}(1 - kh^*) + \rho\bar{g}(I_b + N_{\text{eff}}\alpha f I e^{-c_t h^*}) - 2w(h^* - h_{\text{ref}}) = 0,$$

i.e. 1.1’s fixed-point equation with the toxic baseline-plus-slope amplified by the same N_{eff} . Solve by Newton/bisection on $[0, \text{max_half_spread}]$.

6 Convergence rate of the joint loop

Linear rate. With J_{BR} symmetric and $m_N < 1$, $\|\mathbf{h}^t - \mathbf{h}^*\|_2 \leq m_N^t \|\mathbf{h}^0 - \mathbf{h}^*\|_2$: the joint loop converges linearly at $m_N = N_{\text{eff}}\varepsilon\beta/\gamma$, slowing as N or κ grows. The rounds to tolerance scale like $1/(1 - m_N)$ near the boundary – critical slowing down (§8.3).

Strong-monotonicity view (Narang et al. 2022). The joint Nash gradient operator’s strong-monotonicity constant is the worst-case joint curvature

$$\gamma_{\text{joint}} := \gamma - N_{\text{eff}}\varepsilon\beta, \quad \gamma_{\text{joint}} > 0 \iff \varepsilon < \gamma/(N_{\text{eff}}\beta), \quad (4)$$

which at $\kappa = 1$ is $\gamma_{\text{joint}} = \gamma - N\varepsilon\beta$ (the README’s value, generalising 1.1 §5’s $\gamma_{\text{eff}} = \gamma - \varepsilon\beta$). Under $\gamma_{\text{joint}} > 0$, Narang et al. give $O(1/k)$ on $\mathbb{E}\|\mathbf{h}^k - \mathbf{h}^*\|^2$ for the stochastic loop and linear $(1 - \eta\gamma_{\text{joint}})$ for the deterministic one. When $\gamma_{\text{joint}} \leq 0$ but the weaker *variational-stability* condition holds, the loop still converges, but to a weaker equilibrium concept – a caveat for the systemic reading (§11).

7 The mean-field limit $N \rightarrow \infty$

Strong coupling (κ fixed > 0): systemic collapse. $N_{\text{eff}} = 1 + \kappa(N - 1) \sim \kappa N \rightarrow \infty$, so $\varepsilon^*(N) = \gamma/(N_{\text{eff}}\beta) \sim \gamma/(\kappa N\beta) \rightarrow 0$: *any* $\varepsilon > 0$ destabilises a sufficiently large fully-coupled market.

Mean-field coupling ($\kappa = c/N$): a finite boundary. With each dealer’s footprint $O(1/N)$ (response to the empirical quote distribution), set $\kappa = c/N$ for fixed aggregate intensity $c \geq 0$. Then $N_{\text{eff}} \rightarrow 1 + c$ and

$$\varepsilon < \frac{\gamma}{(1+c)\beta}, \quad \varepsilon_{\text{MF}} = (1+c)\varepsilon, \quad (5)$$

a finite, well-posed continuum boundary: the aggregate attractiveness $\frac{1}{N} \sum_j e^{-c_t h_j^{\text{dep}}}$ concentrates on its population mean (propagation of chaos, Lacker 2016), so the (N, ε) stability region has a clean $N \rightarrow \infty$ limit – the licence to scale Priority 1.5’s phase diagram to a realistic many-dealer market. The two regimes bracket reality; the simulator distinguishes them by the scaling of m_N in N at fixed ε (§10).

8 Corollaries (each a falsifiable prediction)

Corollary 1 (critical dealer count). *At full spillover, the market is stable iff $N < N_c$ with $N_c = \gamma/(\beta\varepsilon) = 1/m_1$. A monopoly-stable market ($m_1 < 1$) is destabilised once the fully-coupled dealer count exceeds $1/m_1$ (e.g. $m_1 = 0.5 \Rightarrow N_c = 2$).*

Corollary 2 (linear-in- N_{eff} modulus). *$m_N = N_{\text{eff}} m_1$ is exactly linear in N_{eff} ; measured joint modulus vs. N (full spillover) is a line of slope m_1 through the origin – a one-parameter check.*

Corollary 3 (critical slowing down). *The relaxation time $\sim 1/(1-m_N)$ diverges as (N, ε) approach the boundary: RRM takes progressively more rounds to settle as $N \rightarrow N_c$, a dynamical early warning of the systemic transition.*

Corollary 4 (spillover is the policy lever). *The boundary depends on N only through $N_{\text{eff}} = 1 + \kappa(N-1)$; fragmenting the informed pool (lowering κ via segmented/tiered RFQ, last-look) re-stabilises a market that “too many dealers” pushed past the boundary.*

As in 1.1§6.4, sweep `toxicity_feedback` f (linear in ε), not α : the multi-dealer phase diagram lives on the (N, ε) plane.

9 Validation protocol and the remaining code task

Predict. For each (N, κ, f) : solve the symmetric PSNE h^* (§5), evaluate m_1 from 1.1’s closed forms at h^* , predict $m_N = N_{\text{eff}} m_1$, the boundary $\varepsilon < \gamma/(N_{\text{eff}}\beta)$, and $N_c = 1/m_1$.

Measure – common-mode probe. Generalise `response_modulus.py` by perturbing all N deployed spreads *in phase* (along the unstable eigenvector $\mathbf{1}$) under common random numbers:

$$m_N^{\text{meas}} = \left| \text{BR}_i(\mathbf{h}^* + \delta\mathbf{1}) - \text{BR}_i(\mathbf{h}^* - \delta\mathbf{1}) \right| / (2\delta) \quad (\text{any } i),$$

which isolates $\lambda_+ = -m_N$. An *anti-phase* probe (eigenvector $\perp \mathbf{1}$) returns $m_1(1 - \kappa)$ and so **identifies κ empirically** from the ratio of the two slopes – a free calibration of the spillover.

Compare – the systemic phase diagram (Priority 3 deliverable). Report m_N^{pred} vs. m_N^{meas} on the (N, ε) grid with cross-seed median + IQR; confirm the linear-in- N law, the integer crossing at N_c , and the critical slowing down. The 2-D (N, ε) stability region is the paper’s systemic-risk figure.

Remaining code task (out of scope here). Add `clients.n_dealers` and `clients.toxic_spillover`, generalise `clients.py` to the coupled τ_i of (1), and add `analysis/multi_dealer_modulus.py` implementing the in-/anti-phase probes (reusing single-dealer machinery). `n_dealers = 1` must reproduce `endo_market.v2` exactly.

10 Honest caveats

- **Full spillover $\kappa = 1$ is the strong assumption.** Real OTC toxic flow is partially segmented ($\kappa < 1$); the honest statement is the interpolated $\varepsilon < \gamma/(N_{\text{eff}}\beta)$ with κ *measured* from the anti-phase probe.
- **Symmetric dealers.** With heterogeneous $(\gamma_i, \varepsilon_i)$, J_{BR} has entries $-\beta$ (spillover) ε_j/γ_i and the spectral radius is dominated by the least-curved (most defensive) dealer – one fragile dealer can set the systemic boundary.
- **Own-franchise benign flow.** Bertrand splitting of uninformed flow is omitted; it adds own- h curvature (raises γ , stabilising) without changing the common-mode-resonance mechanism.
- **Frozen-environment, first-order.** The multi-dealer PerfGD correction (the 1.2×1.3 cross-product) replaces γ_{joint} by an objective curvature and, as in 1.2 §5, can stabilise the loop *beyond* the PSNE boundary; here we establish the blind boundary PerfGD would beat.
- **Cap slack, reference-state constants.** $\bar{g}, \psi, \lambda_q$ and the cap are at the reference state; deep past the boundary the cap re-convexifies and the linearisation (and uniqueness) cease to hold. Report the boundary as a band over the reference-state distribution and over (N, κ) .
- **Mean-field regime selection.** Whether the large-market limit is the collapsing or the finite one depends on the unobserved scaling of toxic capacity with venue count; the simulator can measure which holds, and the paper must not assert one without that measurement.

References

- G. Brown, S. Hod, I. Kalemaj. *Performative Games*. arXiv:2106.09784, 2021. – PSNE existence; product-space contraction (§3–§5). (Lit. #11.)
- A. Narang, E. Faulkner, D. Drusvyatskiy, M. Fazel, L. J. Ratliff. *Multiplayer Performative Prediction: Learning in Decision-Dependent Games*. arXiv:2207.05630, 2022. – joint $O(1/k)$ / variational stability (§6). (Lit. #12.)
- D. Lacker. *A General Characterization of the Mean Field Limit for Stochastic Differential Games*. PTRF, 2016. arXiv:1510.01408. – the $N \rightarrow \infty$ limit of §7. (Lit. #17.)
- J. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative Prediction*. ICML 2020. – the single-agent $m_1 = \varepsilon\beta/\gamma$ boundary that N_{eff} multiplies.
- Builds on 01-analytic-stability-boundary (closed-form $\gamma, \beta, \varepsilon, m_1$) and 02-perfgd-correction (the un-blinding cross-product of §11).